DAY 16 | PRE-READ

When Data Gets Big

A light read before Day 16. About 15 minutes, no code required.

Prepared by **Ahmed** | linkedin.com/in/ahmed-al-imaad

Why read this?

Yesterday you cleaned a few rows of data with pandas on one computer. But what if a company has billions of rows, too many for any single machine? Today you meet Apache Spark, a tool that spreads the work across many computers, and you build a pipeline that could run every night at a real company.

1. Many hands make light work

Imagine counting every vote in a national election alone. It would take years. Instead, thousands of counting centres each count their own share at the same time, then send totals to a central office. Spark works like this. A **driver** (the central office) splits the job, and many **executors** (the counting centres) work in parallel.

2. Plan first, then work

A good chef reads the whole recipe before starting. Spark does the same. When you describe what you want, Spark only writes it down. When you finally ask for a result, it looks at the whole plan, finds the quickest route, and then does the work. This is called **lazy evaluation**.

3. Raw, clean and ready

Layer	Kitchen version	Data version
Bronze	Groceries as delivered	Raw data exactly as received
Silver	Washed and chopped	Cleaned, fixed and de-duplicated
Gold	The finished dish	Summaries ready for reports

4. SQL is a way of asking questions

SQL reads almost like English: "show me the city and total sales, from the orders table, grouped by city, biggest first". Analysts, engineers and many business tools all speak SQL, and Spark does too.

Key idea: a good pipeline can be re-run for any day and always gives the same correct answer. After every join, count the rows. If the number surprises you, stop and find out why.

5. A fast assistant still needs checking NEW (v4)

OpenAI Codex can write a data transformation in seconds. A transformation that looks right can still drop rows or double-count. Today you learn to check AI-written data code against an answer you already know.

6. Think about these before class

- What large task at work could be split among many people working in parallel?
- If a ticket refers to a customer who is not in the customer list, should it appear in the report? Why?
- What checks would you want before an SLA report goes to a customer?

Chiri mo tsumoreba yama to naru (塵も積もれば山となる): even dust, piled up, becomes a mountain